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# When Can Language Models Join Without Lexical Cues?

## Unique-Path Traversal versus End-to-End Accuracy under Path Ambiguity

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<https://github.com/pankajnits/oir>

### Abstract

We ask whether multi-hop answering over knowledge graphs needs readable predicates, or only equality among identifiers. **Opaque isomorphic reasoning (OIR)** is a measurement protocol: a renaming  $\sigma$  of entities and relations that we verify to be injective on each instance’s finite atom set, with no legend in the prompt. HMAC-SHA256 is one generator of  $\sigma$ , not a cryptographic or privacy primitive. We score the answer node, not an emitted  $(r_1, r_2)$ .

**Finding.** Unique sealed routes still join. Competing sealed routes typically yield UNKNOWN, not the decoy. With the start already in  $q$ , OpenAI opaque unique is 32/32; both-opaque unique is 32/32 for Composer 2.5, OpenAI gpt-5.6-sol, and Grok 4.5. On a city-typed OpenAI freeze at  $n=200$  (every HQ is Wikidata P31 city-like; QIDs re-checked live; one draw), unique English/opaque is 200/200, English two-path 199/200, named-arm opaque two-path 38/200 (Wilson [0.14, 0.25]; UNKNOWN 151, decoy 11), and plan 200/200. An  $n=26$  city slice with decoy 0 cannot resolve that  $\sim 5\%$  decoy rate. Exact-match under two sealed routes is strongly sensitive to prompt surface form (Sclar et al., 2024): same-date 512-token case-id wording (ARM line kept) is named-arm 38/200 vs. hashed-id H5-cyclic 134/200; at 4096/medium the ARM line is non-monotonic (cyclic/constant: no ARM 174/151, H5 129/44, K2 179/144). English relation names restore OpenAI with anonymous entities (OE two-path 32/32 vs. both-opaque 1/32). Seals remove lexical cues; they are not a confidentiality proof.

## 1 Introduction

The scored Wikidata question is *What city is the headquarters of the company led by Michael Eisner?* English unique-path context walks `works_at` then `headquartered_in` to Burbank. That success may use English predicates, or only equality among identifiers. The two-path decoy is a second person  $\rightarrow$  company  $\rightarrow$  city route (`partner_of/located_in`) to Kansas City. Opaque two-path keeps the English question and entity names; those predicates become HMAC tokens (`works_at`  $\mapsto$  `E334675310f05` on the scored Eisner item).

**Opaque isomorphic reasoning (OIR)** measures whether a model can join after English predicate strings are replaced by keyed opaque tokens, with no legend in the prompt. A 2-hop factors as two inferred maps; we score only the answer node: selection  $q \mapsto (r_1, r_2)$  and traversal  $(v_0, r_1, r_2, G) \mapsto v_2$ . Opacity does not, by itself, block traversal. The question is when the prompt no longer maps query semantics onto one of two sealed relation sequences. Putting the start entity in  $q$  as the same atom as

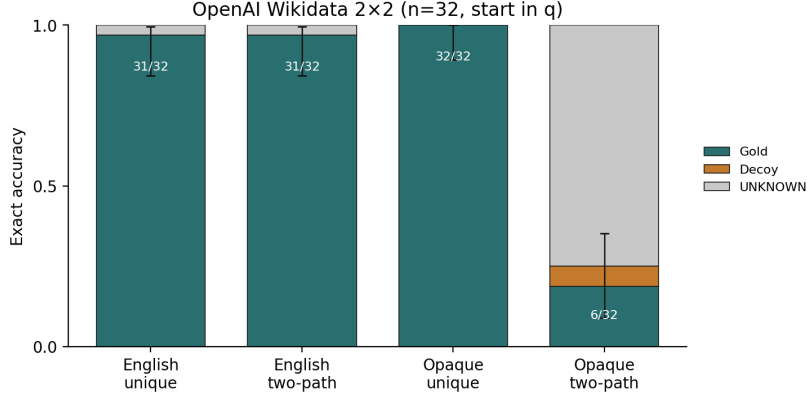


Figure 1: OpenAI WIKI-H5 discovery plot ( $n=32$ , start in  $q$ ). Stacked bars are gold / decoy / UNKNOWN; whiskers are Wilson 95% intervals (one August draw). Named-arm opaque two-path 6/32 (mostly UNKNOWN). City-typed  $n=200$  scores are Section 5.1; prompt-surface ablations are Section 5.2.

in  $V$  (**start in  $q$** ) separates that question from start-entity identification. On the primary  $2 \times 2$ , “lexical cues” means relation strings (entities stay English); both-opaque is a separate ablation.

Figure 1 is the  $n=32$  OpenAI discovery. The powered cell is a city-typed OpenAI  $n=200$  freeze (Section 5.1): unique-path 200/200, named-arm opaque two-path 38/200. This paper reports three measurements. (i) Unique sealed routes still join with the start in  $q$  and no written hop. (ii) Under two competing sealed routes the usual named-arm reply is UNKNOWN, not the decoy; exact-match still moves with prompt surface (Section 5.2). (iii) Matched demonstrations restore shown seals; novel seals do not. What is new is the interaction of opacity with path ambiguity (Section 5.1), and how large a prompt-surface move can be on the same sealed graph (Section 5.2).

## 2 Related Work

Think-on-Graph and Reasoning-on-Graphs select the next readable relation (Sun et al., 2024; Luo et al., 2024). Ciphered ICL is not path ambiguity on a labeled graph (Fang et al., 2025; Guo et al., 2026). GSM-Symbolic swaps entity names in English math (Mirzadeh et al., 2025); 2WikiMultihopQA is a readable multi-hop control (Ho et al., 2020).

ARoG anonymizes *entities* while relations stay the retrieval language (Ning et al., 2026). PrivGemo HMAC-seals entities ( $\phi_Q$ , e.g. `ent_17`); relations stay original labels or clustered schema words, not keyed opaque tokens, and a local Hand answers on the raw graph after a remote Brain ranks retrieved paths (CWQ 67%  $\rightarrow$  59%; WebQSP 75%) (Tan et al., 2026). CoR explores readable relations because entities often have no semantics beyond opaque IDs — the OE hide-set, not relation-only WIKI-H5 (Liu et al., 2026). Tang et al. (2023) find LLMs fail when relation names are replaced by symbols (`r1`) and rules must carry deduction; OpenAI unique-path opaque 32/32 is the complementary case: sealed relations still traverse when topology fixes the route. Guo et al. (2025) report collapse to random guessing among end entities when distractor two-hops have other sources. Named-arm Wiki-H5 opaque two-path, with UNKNOWN allowed, is abstention rather than guessing (Section 5.2). A forced-choice arm is not in this evaluation (Section 7) and would put those protocols on the same scale. OIR withholds retrieved paths on the free arms (Table 1).

Chain-of-thought and Text-to-SQL externalize plans over English (Wei et al., 2022; Yu et al., 2018). We ask a behavioral question: can a pretrained model join when the relation vocabulary is freshly randomized and no plan is given? A sealed plan is then an execution upper bound. NLGraph and Talk like a Graph encode graphs without readable typed predicates (Wang et al., 2023; Fatemi et al., 2024); OIR starts from a labeled KG and removes only the relation lexicon. ARM headers are a prompt-surface feature (Sclar et al., 2024). Symbol tuning, binding-ID analyses, and compositional-limit studies change the labels or the task (Wei et al., 2023; Feng and Steinhardt, 2024; Dziri et al., 2023). OIR holds the join, hop length, graph size, and question template fixed and replaces relation

| Work       | Hidden                                                                                | Remote task                                          | Join under opacity      |
|------------|---------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------|
| ICL        | substitution text                                                                     | ICL task accuracy<br>(bijection inferred)            | ICL tasks               |
| CIPHERS    |                                                                                       |                                                      |                         |
| ARoG       | entities; relations<br>readable; topic<br>names in $q$                                | retrieve + generate<br>(relations, concepts)         | privacy KG-RAG          |
| PrivGemo   | entities (HMAC);<br>relations original or<br>clustered schema<br>words                | rank retrieved paths;<br>local plaintext answer      | privacy KG-RAG          |
| CoR        | opaque entity IDs;<br>relations readable                                              | relation-centric<br>exploration                      | KGQA                    |
| <b>OIR</b> | relations (primary<br>$2 \times 2$ ); entities too<br>(ablation); one join<br>or none | emit graph-namespaces<br>answer (no local<br>decode) | selection vs. traversal |

Table 1: Related protocols. PrivGemo HMAC-seals entities; relations remain original or clustered schema labels. OIR sends one graph and scores the answer token from the same model.

strings; OpenAI unique-path opaque traversal stays at 32/32, so that removal withdraws the cue that would otherwise select between routes rather than blocking execution.

### 3 Problem Formulation

Let  $G = (V, R, T)$  be a labeled multi-relational graph ( $T \subseteq V \times R \times V$ ) and  $q$  a natural-language or programmatic query with answer  $a \in V$ . A **seal**  $\sigma$  is a keyed map from entity and relation strings to opaque tokens that we verify to be injective on the finite normalized atom set of each evaluation instance. We reject any instance where two distinct raw atoms normalize to the same string. Truncated HMAC-SHA256 is not injective over  $\{0, 1\}^*$ ; we use 48-bit truncation because that width is injective on our small per-instance atom sets, not as a cryptographic bound. On the scored Wikidata  $2 \times 2$  instances, no HMAC collision was observed. HMAC is computed from the normalized atom string; entity and relation types are not domain-separated, so identical plaintext yields the same token regardless of type. On each scored evaluation instance the normalized entity strings and relation strings are disjoint, so no entity is identified with a relation. The model receives  $G$  or  $\sigma(G)$  and a query form built from  $\sigma$  (or from English, depending on the arm) and must return the answer node in that namespace; recovering  $\sigma^{-1}$  is not scored. Sealing is deterministic *given the instance key*; the same plaintext need not map to the same token across prompts.

A 2-hop join factors as two maps we measure separately:

$$\text{selection: } (q, G, \mathcal{C}) \mapsto (r_1, r_2), \quad (1)$$

$$\text{traversal: } (v_0, r_1, r_2, G) \mapsto v_2. \quad (2)$$

Traversal is set-valued ( $\tau_G: V \times R \times R \rightarrow 2^V$ ). We evaluate only instances for which  $|\tau_G(v_0, r_1, r_2)| = 1$  for the gold plan (and, on two-path items, for the decoy plan), so each scored route has a unique tail. Uniqueness is per plan, not “one answer in  $G$ .” A sealed join plan supplies  $(r_1, r_2)$  and tests (2). The CEO free-English arm hashes  $q$  piecewise, so  $v_0$  typically does not appear as a single graph atom; it is not a pure test of (1).

A 2-hop is *structurally executable* when  $(r_1, r_2)$  is given, or when topology leaves only one 2-hop from a start already in  $q$ . The prompt *explicitly maps* query semantics onto one relation sequence when it contains English predicates, a legend, or demonstrated seals. When two complete routes exist and relation tokens are arbitrary HMAC strings with no such mapping, selection is *unsupported by explicit cues* under this construction. Residual statistical structure can remain (token form, order, HMAC vs. permutation rates); we do not claim mathematical unidentifiability. Any gold on that arm must use residual cues or other prompt information; low exact-match is not evidence that traversal failed. Dual-path separates the cases: shown seals restore end-to-end accuracy; novel seals do not. The scored output is the answer node or UNKNOWN, not  $(r_1, r_2)$ . We infer whether selection was supported from those outcomes and from controls (unique topology, written hop, matched vs. novel

seals). Throughout, “selection failure” names that inference about the latent  $(r_1, r_2)$  stage; we do not observe the pair. Models may abstain (UNKNOWN) rather than guess (selective prediction; El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). Coverage is the fraction of items with a non-UNKNOWN parse; selective accuracy is exact match among those items, not a forced-choice among the two routes (Kadavath et al., 2022; Wen et al., 2025).

After sealing, the Eisner unique-path CONTEXT is `Michael_Eisner | E334675310f05 | The_Walt_Disney_Company then The_Walt_Disney_Company | Eac04fab538ca | Burbank` (12-hex tokens; entities remain English on the primary  $2 \times 2$ ).

We use five measurement arms. **Readable English** is the baseline (start in  $q$ ; names and predicates unsealed). **Unique opaque** has one 2-hop and no plan: execution when topology uniquely determines the route. **Two-path opaque** has two same-type 2-hops and no plan: end-to-end behavior when two complete routes compete and the prompt supplies no semantic-to-symbol map. **Sealed join plan** writes  $(r_1, r_2)$  over  $\sigma$ -atoms (execution control); a non-LLM router saturates on the same sequence. **Free sealed English** seals the question and graph with no plan and no legend. On CEO, the question is hashed piecewise, so the full person atom in  $G$  typically does not appear in  $q$  (confounded; not the matched factorial). Additional scaffolds (name/key legends, sealed demonstrations) appear as ablations.

## 4 Method

**Seal recipe.** Normalize each atom: strip; collapse whitespace to `_`; map other non-alphanumeric characters (except underscore, period, hyphen) to `_` using the ASCII class `[A-Za-z0-9_.-]` (so `Zürich`  $\rightarrow$  `Z_ri_ch`; pure CJK keeps Unicode); truncate to 64; empty maps to `EMPTY`. The CEO JSON already deleted diacritics upstream. Distinct raw atoms must normalize to distinct strings. Tokens are HMAC-SHA256 over UTF-8 bytes with a per-item 16-byte key; the token is E plus the first 6 digest bytes as 12 hex digits. HMAC is keyed renaming, not a security mechanism: 48-bit truncation is instance-wise injective on our small atom sets. On the scored Wikidata  $2 \times 2$  graphs, normalization and HMAC produced no collisions among prompt atoms. The intervention removes lexical semantics and also changes token-level surface statistics; permutation and alphanumeric generators assess robustness to that latter variation (Table 7). The atom `Michael Eisner` normalizes to `Michael_Eisner`, so those two strings receive the same seal. Hashing the question token-by-token does not reconstruct that graph atom, which is the missing-start failure a name-binding wrapper is designed to prevent. Reported opaque cells are performance under this keyed HMAC construction, not a claim that every opaque identifier distribution yields the same rate.

**Matched  $2 \times 2$  (start in  $q$ ).** The  $n=32$  suite is a seeded random sample of 32 of 73 Wikidata-derived CEO records (`random.Random(20260821).sample`; Vrandečić and Krötzsch, 2014), not a published CEO benchmark and not a hand-picked subset. Each instance is 4 triples (unique-path: 6 entities with a disconnected decoy island; two-path: 5 entities sharing the start). The gold two-hop uses `works_at` then the headquarters relation; the question says “led by”. The competing two-path route is synthetic (`ambig_edges` attaches record  $i+1$ ’s company/HQ under `partner_of/located_in`; not a Wikidata fact; decoy HQ pool = gold HQ pool). To isolate relation selection from start-entity identification we keep the start string in the question and hold hop length, template, and graph size fixed. Two factors vary: relation labels (English vs. HMAC) and topology (one 2-hop from the start vs. two). Tails are intended as the same type; Wikidata P159 was not filtered (buildings/stations/a street among gold HQs; both OpenAI opaque two-path decoys sit on city/non-city pairs; Appendix A). On Wikidata two-path graphs both intermediates are companies (`works_at/headquartered_in` vs. `partner_of/located_in`). In the primary run the gold 2-hop is listed before the decoy 2-hop in CONTEXT (same order in English and opaque). Listing-order robustness is Section 5.2 on the  $n=200$  freeze (same OpenAI tag as the named-arm cell). The  $n=32$  shuffle is a later snapshot (Appendix A), not that comparison.

The primary two-path prompt is a single user message:

```
MODEL UNDER TEST. Read ONLY this file. Use CONTEXT only.
No other files.
Format: ANSWER_PLAIN[...]: <city_or_UNKNOWN>
```

```
ARM OPAQUE_AMBIG: opaque relations, English entities;
two 2-hops from start.
```

It does *not* contain the equality-recipe line (Appendix J). The format token is `<city_or_UNKNOWN>`, but six  $n=32$  gold HQs are not cities (Appendix A). A model that obeys the format on those items must emit UNKNOWN. The powered city-typed cell is  $n=200$  (Section 5.1); the  $n=32$  lock is the three-family pin. OpenAI English two-path is 31/32 under that CONTEXT-only instruction, so the line does not by itself force abstention. Readable names (Eisner, Burbank) still leave OpenAI at UNKNOWN 24/32 on opaque two-path: either parametric HQ is treated as inadmissible, or the verb does not map onto one seal. We cannot separate those readings here (Section 7). The CEO three-arm decoy 2-hop begins at a different person; it is not the matched two-path factorial.

**Further constructions.** The relation-only factorial keeps entities English. A second isolation suite on the same 32 CEO records seals entities, relations, or both, always with the start token in  $q$ . Opaque-entity / English-relation (OE) is the ARoG-like cell: readable predicates, anonymous nodes. Both-opaque (Wiki-OO) seals entities *and* relations; it is a complementary ablation, not the primary relation-only  $2 \times 2$ . Readable entity names therefore leave residual pretrained world knowledge as a possible cue; both-opaque is not a substitute for synthetic identifiers.

WikiMovies facts are sampled from the public MetaQA movie KB (Miller et al., 2016; Zhang et al., 2018). MOVIE-A32 and MOVIE-A100 leave English verbs such as *directed* in  $q$ ; MOVIE-B hashes question atoms except the bracketed start entity. We exclude a genre-vs-language variant because object type would bypass relation selection. This is not an evaluation on the official MetaQA 14K test split. Dual-path holds  $G$  fixed with two complete 2-hop routes and varies whether demonstrations expose the gold seals (Appendix D).

**Scoring.** The format asks for one answer token in the graph’s namespace or UNKNOWN (plain-text city when entities stay English;  $\sigma(a)$  when entities are sealed). Primary metric: exact match (UNKNOWN is a miss). Scored UNKNOWN on Wiki-H5/Wiki-OO is an explicit UNKNOWN line. Dual and some no-hop OpenAI cells stored empty completions as UNKNOWN. On two-path arms we also report decoy and UNKNOWN rates (Table 3). Wilson 95% intervals use  $z=1.96$  (Wilson, 1927) ( $0/32 \Rightarrow [0.00, 0.11]$ ;  $6/32 \Rightarrow [0.09, 0.35]$ ;  $32/32 \Rightarrow [0.89, 1.00]$ ). Those intervals treat the 32 items as Bernoulli trials on a single draw; they are not a generation-variance interval. OpenAI Wiki-H5  $k=3$  (August lock plus two later draws) is Section 5.1. On the OpenAI Wikidata  $2 \times 2$ , English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar exact two-sided complete-case  $p=5.96 \times 10^{-8}$ ; McNemar, 1947). That is the two-path paired contrast, not an interaction test, and it largely measures removing the cue that made one sealed route identifiable. The empirical content is the response mode (UNKNOWN vs. decoy) and prompt-surface packing (Section 5.2). That  $p$  clears 0.001 (confirmatory McNemar only). **Isolation** means one quiz per file with gold answers stored outside the model-visible prompt (stricter than multi-quiz **batch** files). Isolation  $n=32$  uses one quiz per file and a per-item HMAC. Cells marked **n.r.** were not evaluated; we never impute them as zeros. *Mid-hop error* means the model emits the sealed mid-hop company token instead of the headquarters answer.

**Models.** The primary family is OpenAI `gpt-5.6-sol` (Chat Completions; unpinned alias; temperature omitted; 512 completion tokens unless noted). Composer 2.5 and Grok 4.5 used Cursor’s agent SDK on the same quiz files, with a reply-only preamble OpenAI never received. Table 3 is robustness, not a ranking: unique-path opaque still executes on those SDK cells (Composer 27/32, OpenAI 32/32, Grok 30/32). Tools, retry, and empty-reply handling are not equated. The August lock stores the parsed answer, not the API model string. Inference settings and  $k=3$  alias draws are Appendix A. A packaging-only Claude probe is Appendix F. Isolation quizzes and locked scores are at <https://github.com/pankajmits/oir>.

## 5 Results

Table 2 names the cells. OpenAI is primary; Composer/Grok used a different harness (Table 3 is not a ranking). MOVIE-B is a cross-dataset check, not a Wiki-H5 resample. Packaging probes ( $n \leq 16$ ) are Appendix G.

| ID                        | Dataset                | Condition                                             | $n$ | Models           | Scored      |
|---------------------------|------------------------|-------------------------------------------------------|-----|------------------|-------------|
| WIKI-H5                   | Wikidata CEO           | rel-only $2 \times 2$ , start in $q$                  | 32  | OpenAI (primary) | node        |
| WIKI-H5-200               | Wikidata CEO, city P31 | same $2 \times 2$ , QID freeze                        | 200 | OpenAI           | node        |
| Header $\times$ decoy     | same 32 people         | ARM line $\times$ decoy, hashed ids                   | 32  | 3 families       | node        |
| Header $\times$ decoy-200 | same 200 people        | ARM $\times$ decoy, hashed ids                        | 200 | OpenAI           | node        |
| WIKI-OO                   | same graphs            | both-opaque                                           | 32  | 3 families       | $\sigma(a)$ |
| WIKI-OO-200               | $n=200$ freeze         | both-opaque                                           | 200 | OpenAI           | $\sigma(a)$ |
| WIKI-SHUF                 | same two-path          | listing order                                         | 32  | OpenAI           | city        |
| WIKI-SHUF-200             | $n=200$ freeze         | listing order                                         | 200 | OpenAI           | city        |
| MOVIE-A32                 | WikiMovies             | actor $\rightarrow$ movie $\rightarrow$ dir vs writer | 32  | 3 models         | person      |
| MOVIE-A100                | WikiMovies             | same hops; verbs in $q$                               | 100 | OpenAI           | person      |
| MOVIE-A200                | WikiMovies             | $n=100$ prefix + 100 new actors                       | 200 | OpenAI           | person      |
| MOVIE-B                   | WikiMovies             | same hops; verbs hashed                               | 100 | OpenAI           | person      |
| MOVIE-B200                | WikiMovies             | Condition B on $n=200$ freeze                         | 200 | OpenAI           | person      |
| DUAL                      | WikiMovies/synth       | matched vs. novel demos                               | 32  | 3 models         | $\sigma(a)$ |
| DUAL-200                  | WikiMovies $n=200$     | matched vs. novel                                     | 200 | OpenAI           | $\sigma(a)$ |
| CEO-NL                    | Wikidata CEO           | piecewise $q$ , no plan                               | 32  | 3 models         | $\sigma(a)$ |
| CEO-NL-200                | $n=200$ freeze         | piecewise $q$ , no plan                               | 200 | OpenAI           | $\sigma(a)$ |

Table 2: Cell inventory. WIKI-H5-200 is the powered city-typed freeze (OpenAI). Composer/Grok are Table 3, not a ranking. Six  $n=32$  P159 golds are not cities (Appendix A). A100 Condition A uses a thinner ARM than A32. Isolation cells use per-item HMAC.

## 5.1 Opacity $\times$ path ambiguity

WIKI-H5 holds the start string in  $q$  and varies only relation lexicality (English vs. HMAC) and the number of 2-hops from that start (one vs. two). OpenAI is the primary family. Composer 2.5 and Grok 4.5 are Table 3 robustness, not a ranking: they used Cursor’s agent SDK with a preamble OpenAI never received.

**City-typed  $n=200$  (OpenAI).** The  $n=32$  source file has 73 CEO rows and six non-city P159 golds. WIKI-H5-200 is a new freeze: 200 unique people and companies, HQ frequency capped at 3, every HQ an instance of a city-like Wikidata class, person/company/HQ QIDs stored, and all 200 P169+P159+P31 triples re-checked live against the Query Service (`wikidata_ceo_hops_n200.json`). It does not overwrite the  $n=32$  lock. OpenAI named-arm: unique English and unique opaque 200/200, English two-path 199/200, opaque two-path 38/200 (UNKNOWN 151, decoy 11; Wilson [0.14, 0.25]), plan 200/200. The 38/200 rate sits inside the  $n=32$  Wilson band for 6/32. Decoy 0 on the  $n=26$  city slice was a small-sample artifact:  $n=26$  cannot resolve a  $\sim 5\%$  decoy rate. At  $n=200$  the usual reply is still UNKNOWN ( $\approx 14:1$  vs. decoy). Prompt-surface packing of this freeze is Section 5.2.  $n=200$  is OpenAI-only and one draw; three-family cells stay at  $n=32$ .

**$n=32$  discovery.** On unique sealed routes OpenAI executes (opaque unique 32/32; plan 32/32). On competing sealed routes the usual named-arm reply is UNKNOWN, not the decoy. On 26 city-valued items unique English, unique opaque, and English two-path are 26/26; named-arm opaque two-path is 5/26, 2/26, 2/26 across three draws. August: UNKNOWN 21, decoy 0, selective accuracy 5/5. The lock is opaque two-path 6/32 (UNKNOWN 24, decoy 2); both decoys sit on non-city P159 golds. Wilson:  $6/32 \Rightarrow [0.09, 0.35]$ ; city slice  $5/26 \Rightarrow [0.09, 0.38]$ . Figure 1 plots the  $n=32$   $2 \times 2$ .

Let  $\Delta_{\text{opq}}$  be opaque minus English exact-match on  $n=32$ : unique  $+1/32$ , two-path  $-25/32$  (25 English-correct and opaque-wrong, 0 reverse; McNemar  $p=5.96 \times 10^{-8}$ ). Opacity is not a randomized within-item treatment (Appendix I).

**OpenAI  $k=3$ .** Two 15 September draws of the unpinned alias (`gpt56k3r2/r3`) are a new snapshot, not a generation-variance interval. Named-arm opaque two-path is 6/32, 3/32, 3/32; Composer 10, 7, 6 and Grok 1, 0, 0 move the same way. Same-day draws 2 and 3 match; August is highest in all three families. On OpenAI the other named-arm cells hold: unique opaque 32/32, 31/32, 32/32; English two-path 31/32, 30/32, 30/32; plan 32/32 on every draw. Named-arm English two-path is OpenAI 31/32, Composer 26/32, Grok 4/32: a packaging $\times$ family effect, not a ranking. Opaque unique stays high (Composer 27/32, Grok 30/32); a written hop recovers (28/32,

| Wikidata named-arm (iso. $n=32$ ) | OpenAI   | Composer 2.5 | Grok 4.5 |
|-----------------------------------|----------|--------------|----------|
| English, unique                   | 31/0/1/0 | 29/0/2/1     | 32/0/0/0 |
| English, two paths                | 31/0/1/0 | 26/0/6/0     | 4/0/28/0 |
| Opaque, unique                    | 32/0/0/0 | 27/0/3/2     | 30/0/2/0 |
| Opaque, two paths                 | 6/2/24/0 | 10/0/22/0    | 1/0/31/0 |
| Opaque two paths + plan           | 32/0/0/0 | 28/0/3/1     | 32/0/0/0 |

Table 3: Four-way outcomes: gold / decoy / UNKNOWN / other wrong token. OpenAI is the primary family (August lock;  $k=3$  in the surrounding text). Composer 2.5 and Grok 4.5 used Cursor’s agent SDK with a reply-only preamble OpenAI never received; this table is not a ranking. On this table UNKNOWN is an explicit UNKNOWN line (missing/unparsed 0; both-named 0). Composer “other” is truncated gold `Los` (English unique, opaque unique, and plan) plus one wrong city (`Fox_Plaza`→`Los Angeles`). OpenAI’s two decoys are P159 type-asymmetric pairs. Coverage and selective accuracy for every two-path row are in the surrounding text. Grok named-arm English two-path is mostly UNKNOWN.

32/32). Two-path coverage / selective accuracy (named-arm): OpenAI opaque 8/32, 6/8; Composer 10/32, 10/10; Grok 1/32, 1/1. Two other generators still have unique > two-path (Table 7).

## 5.2 Prompt surface as a factor

The named-arm `Wiki-H5` prompt contains this line:

```
ARM OPAQUE_ambiguous: opaque relations, English entities;
two 2-hops from start.
```

That sentence tells the model in English that two 2-hops exist. It is an uncontrolled treatment if abstention is attributed only to opacity-meets-ambiguity. Prompt-surface packing is a factor (Sclar et al., 2024). Two manipulations must not be collapsed.

**Case-id wording, ARM line held fixed.** The case id appears in `Format: ANSWER_PLAIN[id]` and in the ID line. Named-arm uses readable ids (`F22_OPAQUE_ambiguous_i`); the H5-cyclic arm substitutes hashes. ARM line and CONTEXT are identical. On the city-typed  $n=200$  freeze, the same 512-token contrast is 38/200 vs. 134/200 (+96 golds; H5-cyclic has one `NO_OUTPUT`). Coverage moves 49/200 → 155/200 and selective accuracy 38/49 → 134/155: that is not only more guessing. Decoy also rises 11 → 20. The  $n=32$  discovery pair is same-date 11 September: 5/32 vs. 20/32 (+15; one `NO_OUTPUT` each). Both prompts keep the ARM line, so this is not a test of whether the prompt announces ambiguity.  $k=3$  hashed-id gold is 20, 21, 22; against named-arm 6/3/3 only +18 and +19 are same-day.

**ARM line, case ids held hashed.** Table 4 crosses the ARM line with cyclic vs. constant decoys at 4096/medium on the  $n=200$  freeze. H5 cyclic vs. constant McNemar 95 vs. 10,  $p=1.6\times 10^{-18}$ ; no-ARM 30 vs. 7,  $p=1.9\times 10^{-4}$ . The  $n=32$  discovery (Table 8) already showed the H5 interaction (cyclic 20 vs. constant 10; McNemar 14 vs. 4,  $p=0.031$ ) and not no-ARM (gold 25 vs. 27; McNemar 1 vs. 3,  $p=0.625$ ). K2 is a shorter ARM that still names two hops. K2 and no-ARM were not run at 512.

**Listing order.** A shuffled CONTEXT listing of the same  $n=200$  freeze and HMAC keys, same OpenAI tag as the primary named-arm cell, scores opaque two-path 59/200 against 38/200 for the gold-first listing (+21 from reordering). Gold-first is 25/100, decoy-first 34/100; a two-proportion test is  $p\approx 0.16$  (not significant). The  $n=32$  shuffle is a later snapshot (Appendix A) and was too small to read a decoy-first effect (decoy 0/16 vs. 1/16).

**English Grok.** Named-arm English two-path is Grok 4/32 (UNKNOWN 28). The English header×decoy cell (`ENG_NONE_CYC`) is 24/32. That cell has *no* ARM line and hashed ids, so two things differ from the 4/32 named-arm English prompt. Call it English, no ARM, hashed ids—not “arm-neutral English.” A prompt-verbatim Grok rerun of the named-arm English files (`grok45pn`) stays 4/32, so the Cursor preamble is not that gap. OpenAI named-arm English is 31/32 vs. English, no ARM, hashed ids 29/32: OpenAI English does not abstain either way.

| ARM line (OpenAI, hashed ids, 4096/medium, $n=200$ ) | Cyclic  | Constant |
|------------------------------------------------------|---------|----------|
| no ARM                                               | 174/200 | 151/200  |
| H5                                                   | 129/200 | 44/200   |
| K2                                                   | 179/200 | 144/200  |

Table 4: Header $\times$ decoy on the city-typed  $n=200$  freeze (same people as WIKI-H5-200). English, no ARM, cyclic is 197/200. H5 cyclic vs. constant McNemar 95 vs. 10 ( $p=1.6\times 10^{-18}$ ). Same-date 512-token hashed-id H5-cyclic is 134/200 (one NO\_OUTPUT).  $n=32$  golds: Table 8.

| Arm              | Ent.   | Rel.   | OpenAI | Composer | Grok  |
|------------------|--------|--------|--------|----------|-------|
| Unique path      | opaque | Eng.   | 32/32  | 32/32    | 32/32 |
| Two paths        | opaque | Eng.   | 32/32  | 28/32    | 3/32  |
| Unique path      | opaque | opaque | 32/32  | 32/32    | 32/32 |
| Two paths        | opaque | opaque | 1/32   | 0/32     | 0/32  |
| Two paths + plan | opaque | opaque | 32/32  | 28/32    | 32/32 |

Table 5: WIKI-OO and OE on the same 32 graphs (start in  $q$ ). OpenAI OE two-path 32/32; both-opaque two-path 1/32. Plan is a control.

What the data support: exact-match under competing sealed routes is strongly sensitive to prompt surface form. Case-id wording and the ARM line are large, non-exchangeable moves on the same sealed graphs (paragraphs above; Table 4). Budgets differ, so these are not a ranking of manipulations; the ARM effect is not a small add-on to case-id.

A consequence for the contrast with Guo et al. (2025). That work reports collapse to guessing among end entities when distractor two-hops have other sources. Our named-arm OpenAI cell is abstention (UNKNOWN 24/32, decoy 2/32 at  $n=32$ ; UNKNOWN 151, decoy 11 at  $n=200$ ), not a coin-flip onto the decoy. That “UNKNOWN rather than decoy” pattern is not a three-family law. Grok no-ARM opaque commits are 6 gold / 3 decoy cyclic and 1 / 5 constant (7 vs. 8 pooled): chance. For Grok, preferring UNKNOWN over the decoy is a named-arm phenomenon.

Composer/Grok hashed-id H5-cyclic cells used `--preamble none` and score 6/32 and 0/32; the named-arm locks used the Cursor preamble and score 10/32 and 1/32. Both families go *down* with hashed ids, a sign reversal relative to OpenAI. We did not re-query Composer named-arm with the prompt file verbatim, so that reversal mixes case-id wording with preamble deletion and is not interpretable. Grok is at floor on both opaque two-path cells.

### 5.3 Both-opaque and readable relations

Table 5 is a second isolation suite on the same 32 records (ANSWER\_SEALED; start seal in  $q$ ). Unique-path both-opaque is 32/32 for all three families. Composer/Grok relation-only unique (27/32, 30/32 in Table 3) is a different file, not evidence that hiding entities helps. Two-path both-opaque is OpenAI 1/32 (Wilson [0.01, 0.16]), Composer 0/32, Grok 0/32. A written hop recovers (32/32, 28/32, 32/32). OpenAI OE two-path (opaque entities, English relations) is 32/32: readable predicates restore end-to-end accuracy with anonymous nodes under this construction. Grok OE two-path remains 3/32. On the  $n=200$  freeze (OpenAI), unique both-opaque and OE two-path are 200/200; both-opaque two-path is 12/200.

### 5.4 WikiMovies: hashed verbs

Table 6 uses the same actor $\rightarrow$ movie $\rightarrow$ director vs. writer template as MOVIE-A32 (person-valued tails). MOVIE-A32 unique-path with English verbs in  $q$  is 32/32 for all three families; opaque two-path is 17/32, 17/32, 0/32, with plan 32/32. A32’s ARM names two 2-hops; A100’s is OPAQUE\_AMBIG  $n=100$ . On 32 A32 items matched by gold name (31 unique names) OpenAI is 17/32 vs. 29/32 under A100’s thinner ARM. The extra 68 items are 67/68, which is why the  $n=100$  cell is 96/100. Keys differ; they cannot explain an English-name lookup. Director is always the first person-tail, so parametric knowledge and a listing-slot heuristic are not separated. Condition A leaves *directed/starred* in  $q$ ; we do not treat it as a selection result. Condition B hashes those verbs: OpenAI unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; Wilson [0.01, 0.08]).

| WikiMovies people (OpenAI, $n=100$ )          | Unique  | Two-path |
|-----------------------------------------------|---------|----------|
| English relations, Condition A (verb in $q$ ) | 100/100 | 100/100  |
| Opaque relations, Condition A (verb in $q$ )  | 100/100 | 96/100   |
| Opaque relations, Condition B (verbs hashed)  | 100/100 | 3/100    |
| Opaque two-path + plan, Condition B           | —       | 100/100  |

Table 6: MOVIE-A100/MOVIE-B: same actor→movie→director vs. writer template as MOVIE-A32 (Miller et al., 2016; Zhang et al., 2018) (OpenAI  $n=100$ , start in  $q$ ). Condition B: unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; coverage 13/100, selective accuracy 3/13).

Coverage is 13/100 and selective accuracy 3/13. A written hop is 100/100. Composer/Grok Condition A two-path: 62/100, 36/100; Condition B  $n=100$  is OpenAI-only. On the  $n=200$  freeze, Condition A two-path is 122/200 (Wiki-H5 topology ARM, not A100’s ARM OPAQUE\_AMBIG  $n=100$  banner, so not a resample of 96/100); Condition B is 7/200. Movie-B’s ARM names two 2-hops (Wiki-H5-style); A100 Condition A does not. The contrast is a header-width comparison on WikiMovies, not a same-prompt Wiki-H5 replication (Section 5.2). The  $n=32$  MetaQA-style template cell (Appendix H) is not MOVIE-B.

### 5.5 In-prompt seal reuse

Dual-path holds  $G$  fixed with two complete 2-hop routes and varies whether demonstrations expose the gold relation seals. Matched-relation is in-prompt reuse of shown correspondences, not induction of a general cipher. Novel-relation demos teach seals that do not appear on the gold route. An explicit join plan is the execution upper bound. Asymmetric vs. balanced noise: Appendix D.

Isolation  $n=32$  (Figure 2, Appendix D): OpenAI saturates matched-relation under both noise regimes (32/32, decoy 0). Novel-relation balanced is exact match 0/32: all 32 replies are UNKNOWN (24 empty completions). Novel-relation asymmetric is also 0/32 (UNKNOWN 24, of which 17 empty, decoy 8). An explicit dual-path plan scores 32/32. WikiMovies director/writer Dual is 32/32 vs. 0/32 for all three families (dualpath\_wikimovies\_iso\_\*; Composer novel decoy 2/32), with plan 32/32. On the synthetic Dual suite, Composer isolation-tag novel-relation UNKNOWN (composer25) is a sealed-demo helper; free-form composer25ff is decoy 32/32. Grok free-form (grok45ff) novel-relation asymmetric is UNKNOWN on all 32 items. Last-demo answer-token copy is 0/32 on every arm. On the  $n=200$  freeze (OpenAI), matched is 200/200 and novel is 2/200, with plan 200/200.

On a two-path graph a forced uniform guess among the two complete routes is 1/2; that null does not apply once UNKNOWN is a third action. The pattern is consistent with prompt-local reuse of demonstrated seals and provides no evidence of systematic transfer to unseen relation tokens. We do not interpret this as a variable-binding mechanism. Arm definitions: Appendix D.

### 5.6 Missing-start control

The CEO isolation three-arm is an execution upper bound plus a missing-start cell, not the isolation of selection. Join plans score 32/32 on both plaintext and sealed atoms; SealRouter given the gold sealed relation sequence is 32/32; free sealed English scores 0/32 (all UNKNOWN) for Composer 2.5, OpenAI, and Grok 4.5 (Table 10). On the  $n=200$  freeze, plan remains 200/200 and free sealed English 0/200. We exclude a batched free-sealed cell that can succeed by returning a unique neighbor without using  $q$ .

## 6 Discussion

A join is structurally executable when the route is unique or  $(r_1, r_2)$  is given (Section 5.1, Table 5). When two complete routes exist and the prompt supplies no explicit semantic-to-symbol map, the usual named-arm reply is UNKNOWN rather than the decoy (Section 5.1). We do not claim mathematical unidentifiability, and that abstention pattern is not a three-family law (Section 5.2). Exact-match on the same sealed graphs also moves with prompt surface (Section 5.2, Table 4): case-id wording at 512 tokens and the ARM line at 4096/medium are large and non-exchangeable; a listing shuffle of

the same freeze is not significant. English relations restore OpenAI with anonymous entities; Grok still abstains (Table 5). Dual-path reuses demonstrated seals; novel seals do not transfer (Section 5.5). Unique topology with the start in  $q$ , English predicates, matched demonstrations, or a same- $\sigma$  equality recipe restore execution; a bare schema list does not (Table 11). A high English two-path rate does not imply the same join once relations are opaque. OIR diagnoses residual join ability after readable relations and retrieved paths are withheld; PrivGemo keeps those paths and answers locally (Tan et al., 2026).

## 7 Limitations

OpenAI is the primary family. The  $n=200$  freeze is OpenAI-only and one draw; Composer/Grok stay at  $n=32$  (Table 3 is not a ranking). The  $n=26$  decoy-0 slice is underpowered for a  $\sim 5\%$  decoy rate (Section 5.1). The  $n=32$  format token makes six P159 golds ill-posed;  $n=200$  does not use those golds. The  $n=32$  suite is 32 of 73 records, one template, one seed; decoys are cyclic, so Wilson intervals are slightly optimistic (McNemar is not).  $k=3$  mixes the August lock with later alias draws (Section 5.1): same-day draws match, August is high in all three families, and this is not a generation-variance CI. English entities leave parametric cues; HMAC two-path rates vary by generator (6/32, 15/32, 12/32). Selection is inferred from the answer node. Named-arm selective accuracy stays high; Grok no-ARM opaque commits are near chance (Section 5.2). Header $\times$ decoy at  $n=200$  is Table 4; Table 8 is the  $n=32$  discovery. K2 and no-ARM were not run at 512. MOVIE-A32 vs. MOVIE-A100 (17/32 vs. 29/32) is a thinner ARM on WikiMovies;  $n=200$  Condition A uses the Wiki-H5 topology ARM, not that banner. Dual novel OpenAI UNKNOWN includes empty completions. Forced-choice, scored  $(r_1, r_2)$ , and an HQ-city swap are not in this evaluation. The ARM line was not randomized independently of case-id wording on the original lock. Composer hashed-id H5-cyclic used `--preamble none`; the named-arm Composer lock used the Cursor preamble, so that sign reversal is uninterpretable.

## Ethical considerations

Public Wikidata CEO records (CC0; Vrandečić and Krötzsch, 2014) and the WikiMovies/MetaQA graph as released (Miller et al., 2016; Zhang et al., 2018). Appendix cells use 2WikiMultihopQA (Apache-2.0; Ho et al., 2020), Spider (CC-BY-SA-4.0; Yu et al., 2018), and FinQA (Chen et al., 2021). No personally sensitive information. Replies are hash-pinned, not a training dump. OIR is a diagnostic, not a confidentiality mechanism (English names remain on the primary  $2\times 2$ ).

## 8 Conclusion

Unique-route sealed execution succeeds with the start in  $q$  (opaque unique OpenAI 32/32 and 200/200; both-opaque unique 32/32 on three families). On the city-typed  $n=200$  freeze, named-arm opaque two-path is 38/200 (UNKNOWN 151, decoy 11); hashed-id H5-cyclic at 512 tokens is 134/200 (one NO\_OUTPUT). Exact-match under two sealed routes is strongly sensitive to prompt surface: case-id wording moves 96 cyclic golds at 512 tokens; at 4096/medium, ARM on constant decoys moves 107 golds (Table 4). Grok no-ARM opaque commits are near chance, so “UNKNOWN rather than decoy” is not a three-family law.

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## A Protocol, scoring, and models

Quizzes (runs/) and locks (results/): <https://github.com/pankajmits/oir/releases/tag/v1.0.6> (Python  $\geq 3.10$ ; cite tag v1.0.6, not HEAD; shasum -c results/SHA256SUMS). OpenAI: gpt-5.6-sol, no tools, temperature omitted, 512 completion tokens (header $\times$ decoy 4096), reasoning\_effort omitted. August stores the parsed answer, not model/usage;  $k=3$  sidecars store returned\_model. Composer/Grok: Cursor agent SDK and a reply-only preamble OpenAI never received. Cite Dual free-form as \*ff. Empty Dual/no-hop completions were stored as UNKNOWN; Wiki-H5 and Wiki-OO were not. We never impute n.r. as zeros.

Wiki-H5 seed 20260821, 16-byte keys from oir-fact-2x2-iso-v1; 32 of 73 CEO rows; decoys  $i+1 \bmod 32$  (Wilson slightly optimistic). Wiki-H5  $n=32$  isolation is  $32 \times 5$  named arms;  $n=32$  header $\times$ decoy is 224 queries per family;  $n=200$  header $\times$ decoy at 4096/medium is 1400; gpt56n200h512 queried only OPQ\_H5\_CYC. Wiki-OO uses oir-entrel-2x2-iso-v1; CEO-NL seed 20260820. The  $n=12/n=24$  three-arm is **batch** (one shared key oir-ceiling-three-arm-v1); isolation  $n=32$  is per-item keys. Wiki-H5-200 is a separate freeze (wikidata\_ceo\_hops\_n200.json, seed 20260915, city P31, QIDs; SPARQL verify 200/200). HMAC oir-fact-2x2-iso-n200-v1;  $n=32$  prompts are not rewritten. Tags: gpt56n200; header gpt56n200abl; 512 H5-cyclic gpt56n200h512 (gold 134, UNKNOWN 44, decoy 20, other 1, NO\_OUTPUT 1). The  $n=200$  shuffle uses the named-arm tag, not a new snapshot. WikiMovies  $n=200$  keeps the  $n=100$  prefix and adds 100 actors.

Non-city P159 golds: Broadway, NRK\_Marienlyst, Fox\_Plaza, CNN\_Center, Lod\_Railway\_Station; New\_Jersey is a state. OpenAI two-path decoys are items 27 and 31 (both non-city). City slice 5/26 (decoy 0) is underpowered;  $n=200$  is decoy 11. An archived Movie-B ARM banner is 25/100; locked topology-only B is 3/100. CEO free English hashes  $q$  piecewise.

| Opaque generator (OpenAI, start in $q$ ) | Unique | Two-path |
|------------------------------------------|--------|----------|
| Keyed HMAC (E+12hex)                     | 32/32  | 6/32     |
| Random permutation (E+12hex)             | 31/32  | 15/32    |
| Random alphanumeric (not HMAC-shaped)    | 29/32  | 12/32    |

Table 7: Generator robustness on the same 32 Wikidata graphs and question template. Unique-path accuracy exceeds two-path accuracy under three independently generated opaque relation maps; the two-path *rate* is not invariant. Wilson: 29/32  $\Rightarrow$  [0.76, 0.97], 12/32  $\Rightarrow$  [0.23, 0.55], 15/32  $\Rightarrow$  [0.31, 0.64]. Alphanumeric two-path is all abstention or gold (decoy 0).

## B Generator robustness

A random permutation of HMAC-shaped labels on the same graphs preserves unique > two-path (OpenAI 31/32 vs. 15/32; Composer 28/32 vs. 2/32; Grok 31/32 vs. 0/32). An English-entity ambiguity sweep  $k=1..5$  (constant decoy) keeps OpenAI high (29–28/32); Composer falls to 11/32; Grok is 0/32 for  $k \geq 2$ . Disney names remain, so  $k$  is not isolated from entity semantics.

Header $\times$ decoy uses arm-neutral case ids on the same 32 people (header\_decoy\_ablation\_iso\_\*.json; body Section 5.2). Cells are named by header and decoy: H5 keeps the Wiki-H5 ARM line, K2 keeps a shorter two-hop ARM, and “no ARM” deletes the line; cyclic decoys attach record  $i+1$ , constant decoys reuse Cogent Communications  $\rightarrow$  Washington, DC. OpenAI 4096/medium  $n=32$  golds:

| ARM line (OpenAI, hashed ids, 4096/medium) | Cyclic | Constant |
|--------------------------------------------|--------|----------|
| no ARM                                     | 25/32  | 27/32    |
| H5                                         | 20/32  | 10/32    |
| K2                                         | 30/32  | 26/32    |

Table 8: Header $\times$ decoy gold counts on the same 32 people. English, no ARM, cyclic is 29/32. H5 McNemar 14 vs. 4,  $p=0.031$ ; no-ARM gold 25 vs. 27, McNemar 1 vs. 3,  $p=0.625$ . Body Table 4 is the  $n=200$  lock.

Composer/Grok the same day, prompt file verbatim (`--preamble none`): H5 6/2 and 0/0; K2 20/13 and 0/0; no ARM 21/18 and 6/1; English 26 and 24. Named-arm English Grok verbatim (grok45pn) stays 4/32: the Cursor preamble is not that cell. ENG\_NONE\_CYC 24/32 differs in both ARM and case ids (Grok 24/32 vs. no-ARM opaque 6/32; McNemar 18 vs. 0). Grok no-ARM opaque commits are 6/3 cyclic and 1/5 constant (7 vs. 8 pooled). Constant decoy does not raise the Wiki-H5-header cell on any family.

## C Hypothesis status

| Hypothesis                                                                 | Test                                              | Status                                              |
|----------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|
| H1. Opacity alone blocks traversal                                         | unique-path, start in $q$                         | not supported (32/32)                               |
| H2. Written hop restores e2e (control)                                     | two-path + plan; SealRouter                       | control (32/32)                                     |
| H3. Input-binding control (CEO piecewise $q$ )                             | CEO isolation NL                                  | control (0/32); not selection                       |
| H4. Matched demos restore shown seals                                      | dual-path matched vs. novel                       | OpenAI 32/32 vs. 0/32                               |
| H5. End-to-end accuracy degrades when opaque relations meet path ambiguity | matched $2 \times 2$ , start in $q$               | named-arm 25 vs. 0 (city 21 vs. 0); Sec. 5.1        |
| H6. Movie KB, hashed verbs (check)                                         | WikiMovies Cond. B $n=100$                        | OpenAI two-path 3/100                               |
| H7. Readable relations with opaque entities                                | Wikidata OE vs. OO                                | supported for OpenAI here                           |
| H8. Prompt surface moves two-path exact match                              | case-id at 512 cyclic; ARM $\times$ decoy at 4096 | $n=200$ case-id +96; H5 McNemar 95 vs. 10; Sec. 5.2 |

Table 9: Hypothesis status. H4 exact-match is OpenAI Dual; Composer free-form novel is decoy 32/32 (composer25ff). H5 is named-arm WIKI-H5 (OpenAI). H6 is MOVIE-B. H7 is OpenAI-specific (OE vs. WIKI-OO). H8 is prompt surface on the same graphs (Section 5.2).

| CEO-NL isolation $n=32$              | Composer 2.5 | OpenAI | Grok 4.5 |
|--------------------------------------|--------------|--------|----------|
| Scaled plan / SealRouter             | 32/32        | 32/32  | 32/32    |
| Free sealed English (piecewise $q$ ) | 0/32         | 0/32   | 0/32     |

Table 10: Missing-start control, not WIKI-H5. Isolation  $n=32$ . Batch  $n=12/n=24$  uses one shared key (Appendix A).

## D Dual-path arms

We name the arms by what the demos teach. **Matched-relation** (asymmetric or balanced noise): demos expose the gold relation seals; the decoy path uses different seals. **Novel-relation** (asymmetric or balanced noise): demos teach seals that do not appear on the gold route; both routes are novel at quiz time. **Explicit join plan**: gold relation sequence is written in the prompt.

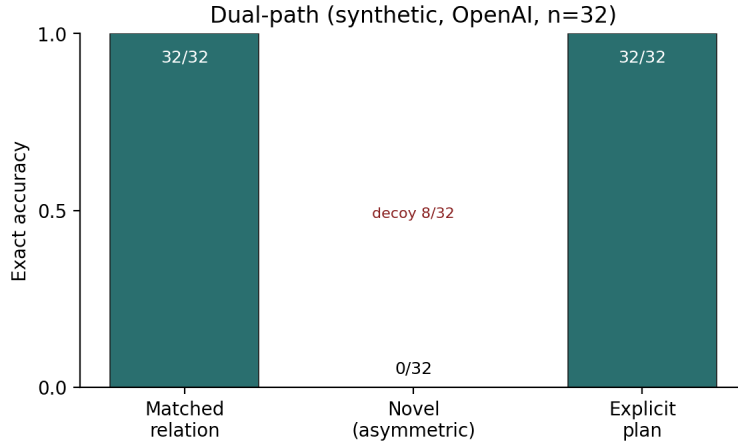


Figure 2: DUAL (synthetic, OpenAI  $n=32$ ): matched 32/32; novel 0/32 (asymmetric decoy 8); explicit plan 32/32.

Isolation  $n=32$  is the reported cell. Cite Dual free-form as composer25ff/grok45ff; the isolation-tag Composer helper is not free-form. Local mistral/qwen35 are not in the roster (plan n.r.).

## E No-hop binding (JSON)

| Arm (no written hop)               | OpenAI | Composer 2.5 | Grok 4.5 |
|------------------------------------|--------|--------------|----------|
| Schema list only                   | 0/32   | 0/32         | 0/32     |
| English protocol (no hop)          | 32/32  | 32/32        | 32/32    |
| Same $\sigma$ , silent             | 6/32   | 19/32        | 31/32    |
| Same $\sigma$ + equality recipe    | 32/32  | 32/32        | 32/32    |
| JSONKEY : match broken             | 0/32   | 0/32         | 0/32     |
| Sealed Q, two routes, recipe       | 31/32  | 25/32        | 32/32    |
| Sealed Q, two routes, vague hints* | 13/32  | 23/32        | 31/32    |

Table 11: Selection support without a written hop (Wikidata CEO JSON; city or mentor country is the distractor; isolation; per-item HMAC). Schema-only and broken token overlap fail for all three. An English protocol or an equality recipe that does not name the field saturates. Silent same- $\sigma$  overlap and vague hints are family-dependent, not a three-family restore. OpenAI silent 6/32 and vague-hint 13/32 include 24 and 15 empty completions stored as UNKNOWN. \*Composer vague-hint distractor 1/32. JOIN was not evaluated at  $n=32$ . Wilson: 32/32  $\Rightarrow$  [0.89, 1.00], 0/32  $\Rightarrow$  [0.00, 0.11], 6/32  $\Rightarrow$  [0.09, 0.35], 31/32  $\Rightarrow$  [0.84, 0.99]. Silent same- $\sigma$  OpenAI 6/32 is a JSON-binding cell, distinct from WIKI-H5. NS\_SILENT ARM lines say “Match impossible”; those 0/32 rows are not a sealed-join collapse.

Prompts are fixed English blocks, not WIKI-H5. Equality means token identity (Appendix J). English JSON keys alone do not restore the walk (1/6).

## F Sealing middleware

Bind a typed full name to one graph atom, then seal. Isolation without binding is 0/6 vs. layer 6/6 (Composer, OpenAI, unversioned Claude; plaintext in the packed prompt: 0). Those files attach a join: an execution control, not the free WIKI-H5 6/32 cell.

## G Packaging probes

Not the matched  $2 \times 2$ . Free sealed English 0/12 vs. plan 12/12 on table/JSON/vault (Composer, OpenAI). Spider identifier-in- $q$   $n=100$ : EQ 51/100, namespace-break 0/100 (OpenAI; not official EM). If the identifier is not in  $q$  ( $n=26$ ), the namespace-break cell is 20/26: an English  $q$  plus legend can still bind. Excel NL names both predicates: Composer 0/16 vs. plan 16/16. FinQA NL 0/8 vs. gold programs 8/8 (Chen et al., 2021). 2Wiki-CF: Composer plain NL 200/200; OpenAI 150/200 with overlap leak 50/200 (not a 2Wiki eval). Long context: NL 0/8, plans 8/8. Rename under a second HMAC: 12/12. SealRouter and gold-SQL are non-LLM ceilings (Figure 3).

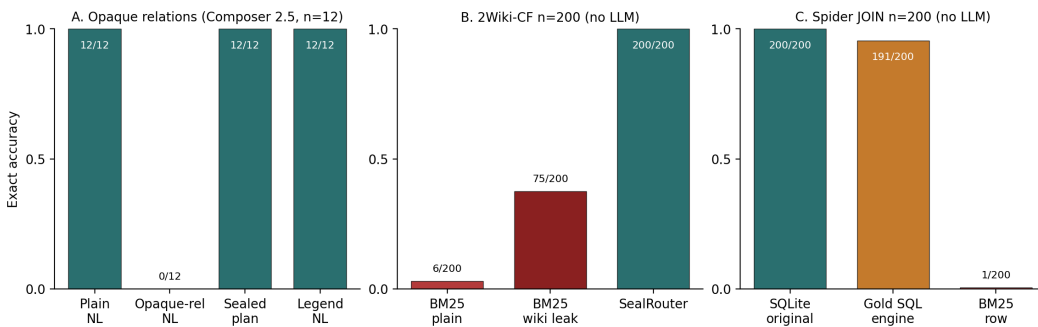


Figure 3: Non-LLM and packaging controls. Left: Composer  $n=12$  (plain 12/12, opaque-rel NL 0/12, plan 12/12). Middle: 2Wiki-CF BM25 6/200, Wikipedia leak 75/200, SealRouter 200/200. Right: Spider sqlite 200/200, gold-SQL 191/200, BM25 1/200.

## H Extended scores

MetaQA-style 2-hop templates on the same public movie graph are a question-form control, not an evaluation on the official MetaQA 14K test split. Plaintext is 32/32 for all three families. Condition A (English question over opaque relations) is OpenAI 32/32, Composer 19/32, Grok 15/32. Condition B hashes those verbs except `[start]`: OpenAI 25/32, Composer 27/32, Grok 14/32 (plans 32/32, 31/32, 32/32). That  $n=32$  template cell is not MOVIE-B.

## I Sign-flip of the H5 interaction

Let  $d_i = \mathbf{1}[\text{opaque correct}] - \mathbf{1}[\text{English correct}]$  and  $c_i = d_i^{\text{two}} - d_i^{\text{unique}}$ . Of 32 WIKI-H5 items, 24 have  $c_i = -1$ , one has  $c_i = -2$ , 7 have  $c_i = 0$ ;  $T = -26$ . Under exchangeability of unique-path and two-path opacity effects,  $p = 2^{-24} = 5.96 \times 10^{-8}$ . That  $p$  coincides numerically with the two-path McNemar  $p$  (25 vs. 0; McNemar, 1947); the tests are not the same. Opacity is not a randomized within-item treatment. The body reports Wilson intervals and the paired counts.

## J Equality recipe (JSON cell)

Verbatim protocol used in Table 11, *not* in WIKI-H5: “QUESTION and CONTEXT use the same encoding. Identical tokens are the same entity or the same field (equality, not English). START is a QUESTION token that also appears as an id value in CONTEXT. The asked field is a QUESTION token that also appears as a JSON key. Walk CONTEXT by token equality. If more than one reading is possible, UNKNOWN.”

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